

BOBOC AIR BASE, Romania

WMO: 150001

Lat: 45.210N

Lon: 26.972E

Elev: 338

StdP: 14.52

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.4	12.6	2.8	6.4	10.9	6.9	7.9	13.8	42.7	17.8	34.2	25.4	11.0	20	0.571

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	22.7	92.3	69.6	89.7	68.7	87.5	67.8	72.7	86.6	71.0	84.5	69.6	83.3	11.0	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.1	104.4	79.9	66.3	98.0	77.9	64.6	92.3	76.3	36.4	86.5	35.1	84.7	33.9	83.5	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
29.5	24.3	22.4		DB	3.0	96.8	7.5	3.3	-2.4	99.2	-6.8	101.1	-11.0	103.0	-16.4	105.4
				WB	3.6	75.2	7.3	1.8	-1.6	76.5	-5.9	77.6	-10.0	78.6	-15.3	79.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	53.2	29.1	32.7	43.3	53.5	63.2	70.1	73.9	74.6	65.1	53.0	44.1	34.2	(6)
(7)		DBStd	17.20	9.12	9.90	8.09	6.37	5.26	5.14	4.44	4.97	6.23	7.25	7.74	7.66	(7)
(8)		HDD50	2153	648	487	232	33	0	0	0	0	0	54	210	489	(8)
(9)		HDD65	5192	1113	905	674	351	97	14	1	2	76	377	628	954	(9)
(10)		CDD50	3311	0	2	23	137	409	602	742	763	455	146	32	0	(10)
(11)		CDD65	873	0	0	0	5	40	166	278	300	81	3	0	0	(11)
(12)		CDH74	9074	0	0	0	103	449	1568	2736	3304	867	46	1	0	(12)
(13)		CDH80	3363	0	0	0	19	62	493	1088	1451	248	2	0	0	(13)
(14)	Wind	WSAvg	10.4	11.0	12.0	12.4	11.4	10.2	9.7	9.1	9.7	10.0	9.9	9.4	10.1	(14)
(15)	Precipitation	PrecAvg	21.1	1.0	0.7	1.1	1.7	2.6	3.0	2.6	2.0	2.0	1.8	1.3	1.3	(15)
(16)		PrecMax	30.0	2.4	2.2	2.4	4.3	7.7	4.6	6.7	5.1	4.6	5.2	3.0	3.5	(16)
(17)		PrecMin	13.5	0.1	0.1	0.0	0.7	0.7	0.9	0.3	0.4	0.2	0.1	0.0	0.0	(17)
(18)		PrecStd	4.4	0.6	0.6	0.6	0.8	1.6	1.0	1.7	1.3	1.2	1.2	0.9	0.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.9	60.5	71.4	82.8	84.5	91.8	95.3	98.2	89.7	78.9	70.0	57.4	(19)
(20)			MCWB	44.2	52.9	53.4	61.7	63.3	71.4	69.8	70.2	65.5	60.5	57.3	49.2	(20)
(21)		2%	DB	46.8	53.8	66.1	76.9	81.0	88.2	91.7	93.2	86.2	74.8	64.4	51.9	(21)
(22)			MCWB	42.2	47.1	51.0	59.9	63.5	69.7	69.6	69.3	64.2	60.4	55.7	46.1	(22)
(23)		5%	DB	43.1	48.6	61.8	71.6	78.8	85.9	89.3	89.9	82.5	70.1	59.3	48.0	(23)
(24)			MCWB	40.7	43.6	49.3	56.8	62.7	68.9	69.3	68.2	63.0	58.0	53.1	43.3	(24)
(25)		10%	DB	40.8	44.7	57.1	67.6	75.6	82.6	86.0	87.8	79.0	66.0	55.5	44.5	(25)
(26)			MCWB	38.7	41.5	47.2	54.9	61.4	67.4	68.7	67.2	61.4	56.1	51.0	41.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.4	52.6	55.4	63.3	67.6	74.4	73.7	74.6	66.8	64.3	59.0	51.2	(27)
(28)			MCDWB	50.3	59.7	67.3	79.3	78.4	87.7	87.8	87.7	81.7	73.3	67.7	55.5	(28)
(29)		2%	WB	42.7	48.2	52.1	60.5	65.6	72.3	72.3	72.6	65.5	60.8	56.2	47.1	(29)
(30)			MCDWB	44.4	53.5	63.9	75.0	77.3	85.1	85.8	87.6	81.6	71.1	62.8	51.0	(30)
(31)		5%	WB	40.7	44.2	50.1	58.0	63.9	70.2	70.9	70.2	64.1	58.8	54.1	44.1	(31)
(32)			MCDWB	42.7	47.2	60.2	70.4	75.5	82.4	84.2	85.9	80.0	69.2	59.2	46.2	(32)
(33)		10%	WB	38.5	41.5	47.7	55.5	62.3	68.5	69.7	68.8	62.7	56.8	51.7	41.5	(33)
(34)			MCDWB	40.2	44.4	56.1	64.6	73.5	80.0	82.8	84.6	77.2	64.7	55.3	44.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	12.4	13.2	18.3	21.3	20.9	20.2	21.0	22.7	21.7	18.4	14.4	12.8	(35)
(36)			MCDBR	16.2	20.7	27.9	30.3	26.0	24.7	25.5	26.3	27.4	24.9	21.6	18.9	(36)
(37)			MCWBR	12.4	14.8	16.2	16.2	12.5	11.2	9.7	9.8	11.2	13.4	13.8	14.0	(37)
(38)		5% WB	MCDBR	15.2	18.8	25.2	27.1	22.9	21.8	22.1	23.7	25.5	24.0	19.5	16.9	(38)
(39)			MCWBR	12.0	14.3	15.4	15.0	12.2	10.6	9.9	9.8	11.2	13.7	12.7	13.4	(39)
(40)																
(41)	Clear-Sky Solar Irradiance	taub		0.332	0.362	0.396	0.455	0.429	0.431	0.446	0.460	0.421	0.384	0.357	0.333	(40)
(42)		taud		2.332	2.249	2.208	2.063	2.218	2.246	2.211	2.160	2.234	2.320	2.348	2.343	(41)
(43)		Ebn at Noon		242	255	262	255	267	266	261	251	251	244	229	227	(42)
(43)		Edh at Noon		25	33	39	49	44	43	44	44	38	30	25	23	(43)
(44)	All-Sky Solar Radiation	RadAvg		415	677	1046	1438	1810	1929	1962	1768	1284	832	456	349	(44)
(45)		RadStd		54	86	109	148	141	141	138	114	108	94	47	45	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (59 neighbors)	+0.88	N/A	N/A	N/A	N/A	N/A	N/A	-247	+126	+68	

Nomenclature: See separate page